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#!/usr/bin/env bash
#
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# /home/azureuser/AGY-
MASTER/DRAFTS/03-immutable-files.sh
(DRAFT ONLY)
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# Purpose: Applies Linux Inode
Immutability (chattr +i) to critical
#          configuration files, and
LAST, to the AGY-MASTER directory
itself.
#
# CHANGE LOG (fixed from previous
draft):
# - FIXED ORDERING BUG: the previous
version applied chattr +i to the
#   AGY-MASTER directory without
first guaranteeing all runtime
subdirectories
#   (TASKS, INCIDENTS, ACTIONS,
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REPORTS, DRAFTS) already existed.
Locking the
#   parent BEFORE those
subdirectories exist would make it
impossible to
#   create them afterward without a
chattr-capable sudo session, which
draft
#   01 deliberately removes. This
script now creates and verifies all
#   subdirectories FIRST, and locks
AGY-MASTER itself LAST.
# - Added existence + ownership
verification before locking.
#
# ARCHITECTURAL SECURITY COUPLING
(WHY THIS WORKS):
# -----
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# In the original audit, 'chattr +i'
was classified as "bypassable"
because:
#   AGY -> child process -> sudo ->
root -> chattr -i <file>
#
# Once draft 01 (01-sudoers-
restricted.draft) is active:
# 1. 'chattr' is STRICTLY EXCLUDED
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from sudoers (deny-by-default).
# 2. 'azureuser' has CapEff = 0 (zero
effective Linux capabilities).
# 3. Removing '+i' requires the
kernel capability
CAP_LINUX_IMMUTABLE, which
#   is only available to a real root
context – not grantable via this
#   restricted sudoers file.
# 4. Without sudo access to chattr
and without CAP_LINUX_IMMUTABLE,
azureuser
#   (even a compromised AGY process
running as azureuser) cannot remove
'+i'
#   under any circumstance – not via
C, Python, or direct syscalls.
# 5. Direct syscalls (unlink, rename,
truncate, or the ioctl
FS_IOC_SETFLAGS
#   used by chattr itself) will
return errno EPERM (Operation not
permitted)
#   directly from the ext4
filesystem layer, before reaching any
userland code.
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set -euo pipefail
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echo "=== [DRAFT EXECUTION] STEP 1:  
ENSURING RUNTIME SUBDIRECTORIES EXIST  
==="
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echo "(Must happen BEFORE locking the  
parent, or these can never be created  
later)"
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REQUIRED_SUBDIRS=(  
    "/home/azureuser/AGY-  
MASTER/TASKS/queue"  
    "/home/azureuser/AGY-  
MASTER/TASKS/active"  
    "/home/azureuser/AGY-  
MASTER/TASKS/completed"  
    "/home/azureuser/AGY-  
MASTER/INCIDENTS/active"  
    "/home/azureuser/AGY-  
MASTER/INCIDENTS/resolved"  
    "/home/azureuser/AGY-  
MASTER/ACTIONS/pending"  
    "/home/azureuser/AGY-  
MASTER/ACTIONS/approved"  
    "/home/azureuser/AGY-  
MASTER/ACTIONS/history"  
    "/home/azureuser/AGY-
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MASTER/REPORTS/daily"
    "/home/azureuser/AGY-
MASTER/DRAFTS"
)

for dir in "${REQUIRED_SUBDIRS[@]}";
do
    mkdir -p "$dir"
    chown azureuser:azureuser "$dir"
    echo "[+] Verified: $dir"
done

echo "=== STEP 2: APPLYING
IMMUTABILITY TO CRITICAL TOP-LEVEL
FILES ==="

PROTECTED_FILES=(
    "/home/azureuser/AGY-
MASTER/AGENTS.md"
    "/home/azureuser/AGY-
MASTER/PROJECT_REGISTRY.md"
    "/home/azureuser/AGY-
MASTER/POLICIES/safety_levels.json"
    "/home/azureuser/AGY-
MASTER/POLICIES/protected_registry.js
on"

"/home/azureuser/.gemini/config/hooks
.json"
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"/home/azureuser/.bashrc"
"/home/azureuser/.profile"
)

for file in "${PROTECTED_FILES[@]}";
do
    if [[ -f "$file" ]]; then
        echo "[+] Setting immutable
flag (+i) on: $file"
        sudo chattr +i "$file"
    else
        echo "[-] File not found,
skipping: $file"
    fi
done

echo "=== STEP 3 (LAST): LOCKING AGY-
MASTER DIRECTORY ITSELF ==="
echo "(All subdirectories above must
already exist – verified in Step 1)"

if [[ -d "/home/azureuser/AGY-MASTER"
]]; then
    echo "[+] Setting directory-level
immutable flag on:
/home/azureuser/AGY-MASTER"
    sudo chattr +i
"/home/azureuser/AGY-MASTER"
else

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        echo "[!] ERROR:
/home/azureuser/AGY-MASTER does not
exist. Aborting."
        exit 1
    fi

    echo "=== IMMUTABILITY STATUS
INSPECTION ==="
    for file in "${PROTECTED_FILES[@]}";
    do
        if [[ -f "$file" ]]; then
            lsattr "$file"
        fi
    done
    lsattr -d "/home/azureuser/AGY-
MASTER"

    echo "=== SANITY CHECK: CONFIRM
SUBDIRECTORIES ARE STILL WRITABLE
INSIDE THEMSELVES ==="
    TEST_FILE="/home/azureuser/AGY-
MASTER/TASKS/queue/.write-test-$$"
    if touch "$TEST_FILE" 2>/dev/null;
    then
        echo "[+] PASS: Can still write
inside TASKS/queue/ (expected – only
the parent is locked)"
        rm -f "$TEST_FILE"
    else
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```
    echo "[!] WARNING: Cannot write  
inside TASKS/queue/ – check  
ownership/permissions"  
fi
```

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echo "=== COMPLETED DRAFT  
VERIFICATION ==="
```